



Lithium-ion battery aging mechanisms and life model under different charging stresses

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HIGHLIGHTS

- Battery aging mechanisms under different charging stresses are identified.
- When less than 1C, Charging current is the deciding factor of active material loss.
- When less than 4.2 V, Cut-off voltage is the deciding factor of lithium loss.
- A capacity degradation rate model at different aging states is established.
- The aging stage when reducing charging stresses is found to delay battery aging.

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ABSTRACT

The charging time-consuming and lifespan of lithium-ion batteries have always been the bottleneck for the tremendous application of electric vehicles. In this paper, cycle life tests are conducted to reveal the influence of different charging current rates and cut-off voltages on the aging mechanism of batteries. The long-term effects of charging current rates and cut-off voltages on capacity degradation and resistance increase are compared. The results show that there exists a critical charging current and a critical charging cut-off voltage. When the charging stress exceeds the critical value, battery degradation speed will be greatly accelerated. Furthermore, battery aging mechanisms at various charging currents and cut-off voltages are investigated using incremental capacity analysis. It is indicated that charging current and cut-off voltage should be reduced to retard battery degradation when the battery degrades to a certain extent. The time when the loss of electrode material accelerates is taken as the crisis to reduce charging current and the time when the loss of lithium inventory accelerates is taken as the crisis to reduce charging cut-off voltage. Moreover, an experiential model quantitatively describing the relationship between capacity degradation rate and charging stresses at different aging states is established.

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1. Introduction

In recent years, due to the excellent properties including high power and energy densities, broad operating temperature range, long cycle life, no memory effect and low self-discharge rate [1,2], lithium-ion batteries have been considered as the most promising power source for electric vehicles (EVs), hybrid electric vehicles (HEVs), portable electronics and energy storage system. There has

been an increasing interest in the fast charging capability of lithium-ion batteries because of the need for convenience in practical applications. Reducing charging time is crucial to increase the marketability of EV. However, fast charging typically involves high charging current rates, which may accelerate battery degradation [3,4]. Fast charging is conflict with extending the lifespan of lithium ion battery to mitigate the high cost. Hence, it becomes necessary to identify the battery aging mechanisms and quantify the effects that different charging stresses introduce to the battery.

A high-quality charging pattern of lithium-ion battery will achieve the balance between the charging speed and battery lifespan. Numerous charging strategies aiming at increasing charging

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speed, enhancing charging performance and maximizing battery life have been reported in literature. Based on the electrochemical properties of lithium-ion battery, Ref [5,6] optimize charging strategy to restrain the side reactions. Those strategies can help to reduce battery degradation but can barely be implemented in real applications. Based on the equivalent circuit model, Ref [7–9] develop dual-objective optimal charging strategies, and the adjustments among charging speed, charging efficiency and cycle life can be realized. These strategies are easy to implement but lack the support from electrochemical mechanisms. Ref [10–13] propose a multi-stage constant current (CC) charging strategy and global optimization techniques are utilized to obtain the optimal charging pattern covering the whole charging process. The long-term effects of multistage fast charging on LiFePO₄-based battery are investigated in Ref. [14]. Based on the battery aging process, several works compare and evaluate different charging strategies [15,16] and charging stresses including charging current and charging cut-off voltage [17], but they have not involved the aging mechanisms of lithium-ion battery under different charging currents and cut-off voltages.

Most of charging strategies in literature lack of theoretical guidance in the selection of charging current and charging cut-off voltage. It is because that lithium-ion battery aging mechanisms under different charging current rates and cut-off voltages are not clear, and a quantitative model that describes the relationship between capacity degradation speed and charging stresses has not been established.

There are some literature researching battery degradation under different operating conditions, such as ambient temperature, SOC range, charging/discharging rate and cut-off voltage [18–21]. Battery lifespan can be extended by optimizing the operating conditions. The optimization potential for discharging processes is usually very limited, as the discharging conditions depend largely on the users' usage habits. However, for the purpose of retarding battery degradation, the charging process can be influenced substantially by adjusting charging current profile and cut-off voltage [22]. Ref [23] reveals that when the charging current rate and cut-off voltage exceed a certain limit at low temperature, the main aging mechanism of LiFePO₄ battery is lithium deposition, but it may be different at normal temperature. Ref [24] investigates battery aging behaviors during long-term cycling at high charging rates in order to elucidate the aging effects of lithium plating. Ref [25] reveal the effects of charging rates on the graphite electrode of commercial lithium-ion cells by post-mortem study, and the results indicate that higher charging currents increase the quantity of negative electrode surface film. Ref [26,27] demonstrate that the formation and development of solid electrolyte interface (SEI) film at the anode and other side reactions will consume active lithium, which is the main cause of battery capacity degradation. Several studies regarding the degradation mechanisms of lithium-ion batteries agree on that the loss of lithium inventory (LLI), loss of active material (LAM) and reaction kinetics degradation are the main causes of battery aging [15,28]. Incremental capacity analysis (ICA) is an effective, in-suit and non-invasive electrochemical analysis method to identify battery aging mechanisms with high fidelity and accuracy, and it has been widely used in literature [29–31]. Battery life model is very important for battery health management and life prediction, which are the key issues of battery management system (BMS) [32]. By adopting the general power law equation described by Ref. [33], Ref [34] develops a semi-empirical life model that considers the effects of time, temperature, depth of discharge and discharge rate. However, the model does not describe the relationship between capacity degradation speed and charging stresses.

1.1. Contributions of the paper

In this paper, cycle life tests on commercial lithium-ion batteries are conducted to reveal the aging mechanisms of battery under different charging currents and charging cut-off voltages. The long-term effects of charging stresses on capacity degradation and resistance increase are compared. Furthermore, The ICA is used to identify the aging mechanisms. The main contributions of this paper are summarized as:

1. Identifying battery aging mechanisms under different charging stresses. When lower than the critical value, charging current is the deciding factor of LAM while its effects on the amount of LLI are not significant. On the contrary, charging cut-off voltage is the deciding factor of LLI while it has no effect on LAM. When the charging current or cut-off voltage is higher than the critical value, the battery will degrade quickly, at the same time, both LLI and LAM will accelerate.
2. Finding the aging stage to reduce charging stresses. The LAM and LLI will accelerate when the battery degrades to a certain extent, especially for the battery charged at critical values. Therefore, reducing charging stresses at the appropriate aging stage is necessary to delay battery aging. The time when the LAM accelerates is taken as the crisis to reduce charging current and the time when the LLI accelerates is taken as the crisis to reduce charging cut-off voltage.
3. Establishing capacity degradation rate model. Based on the experiential formula of battery cycle life, a quantitative model describing the relationship between capacity degradation rate and charging stress at different aging states is established.

1.2. Organization of the paper

The rest of this paper is organized as follows. Section 2 introduces the cycle life tests of battery under different charging stresses. Section 3 presents the test results of battery aging in detail. Section 4 discusses the results in a deeper manner by analyzing battery degradation mechanisms and proposing a quantitative capacity degradation rate model. In Section 5, we summarize the conclusions of this paper.

2. Experimental

To investigate the aging mechanisms of lithium-ion battery and establish life degradation model under different charging stresses, cycle life tests were conducted under different conditions including varied charging current rates and cut-off voltages, and the reference performance test (RPT) which was developed to access the basic performance of lithium-ion battery was performed periodically. The tested batteries are commercial 18650 batteries whose parameters are listed in Table 1. The batteries were subjected to various cycle aging conditions by using a multichannel Arbin battery tester, and in the process of cycle life test, the batteries were

Table 1
Nominal specifications of the battery.

Item	Specification
Cathode material	LiCoO ₂
Anode material	Graphite
Nominal capacity	2.4Ah
Max. charging current	1C
Charging cut-off voltage	4.20 ± 0.05 V
Max. discharging current	3C
Discharging cut-off voltage	3 V

placed in an environment chamber to maintain a constant ambient temperature of 25 °C.

2.1. Reference performance test

We conducted the RPT on 21 batteries from the same batch before cycle life tests. The RPT comprised a capacity test, 1/20C rate test and hybrid pulse power characterization (HPPC) test. Before the RPT, the new batteries were stabilized by 5 charging/discharge cycles at 1/3C rate. The capacity was measured by 3 cycles comprising a 1/2C rate constant current–constant voltage (CC–CV) charging and a 1/2C rate constant current discharging, which was recommended by the manufacturer. The 1/20C rate charging/discharging test with a 2-h rest period between the regimes was used to determine the maximum available capacity and calculate the incremental capacity (IC) curve and the OCV–SOC curve. The objective of the HPPC test was to measure the resistance at different state of charge (SOC).

2.2. Cycle life tests

Cycle life tests under 7 charging stresses were performed on 21 batteries to analyze their impact on the aging mechanisms, and 3 batteries were randomly allotted to each charging stress, which is listed in Table 2. In the cycle life tests, the discharging cut-off voltage was 3 V and the discharging current rate was set as 1C to balance the experiment time and the effects of discharging process on battery degradation. In consideration of the practical application of lithium-ion batteries in EV, battery packs are charged by a multistage reduction current after the battery voltage reaches the charging cut-off voltage. Therefore, the charging pattern in the cycle life test was constant current charging to the cut-off voltage followed by a multistage reduction current charging. When the charging current reduced to 1/20C and at the same time, the battery voltage reached to the cut-off voltage, the charging process was terminated. The RPT was conducted after the battery suffered a certain number of cycles, which was aimed to obtain the changing tendency of battery parameters along with aging.

3. Results

3.1. Battery initial characteristics

When a full-charged battery is discharged by a 1/20C rate current, the potential drop of polarization and ohmic are very small, the terminal voltage approaches to the equilibrium potential, so the discharging capacity at 1/20C rate is approximately equal to the maximum available capacity of the battery, which is decided by the amount of electrode active material. Fig. 1 displays the distribution of the battery initial maximum available capacity and resistance that were measured from the PRT. Fig. 1 (a) shows that the capacities are in the range of 2.51Ah–2.55Ah. The curve in the histogram

represents the probability distribution of initial capacity. Fig. 1 (b) compares the distribution of the capacities with a set of standard normally distributed data, and the result indicates that the initial capacity is normally distributed with 95% confidence intervals. Fig. 1 (c) shows that the resistances of the tested batteries at 40% SOC are in the range of 0.031Ω–0.036Ω, and Fig. 1 (d) indicates that the resistance is normally distributed with 95% confidence intervals. The 21 batteries were divided into 7 groups randomly for cycle life test, and the small variations and normal distributions of the battery initial parameters indicate that the tested batteries are not likely to include any outliers and the results of cycle life tests among the batteries are comparable.

3.2. Battery maximum available capacity degradation

To investigate the effects of different charging stresses on battery cycle life, we compared the maximum available capacity degradation along with cycles under varied charging current rates and cut-off voltages, which is illustrated in Fig. 2. To exclude the effects of random factors, the mean capacity of three tested batteries is used for comparison. Fig. 2 (a) displays the long-term capacity degradation of the tested batteries under different charging current rates but the same cut-off voltage that is 4.2 V. For 0.5C rate charging, the capacity degradation rate is 0.020% per cycle before 150th cycle, and during 150th cycle to 800th cycle, it is stable at 0.0156%, while after 800 cycles, it is 0.0214%. For 0.8C rate charging, the capacity degradation rate is 0.0243% per cycle before 150th cycle, and during 150th cycle to 800th cycle, it is stable at 0.0175%, while after 800 cycles, it is 0.0209%. For 1C charging rate, the capacity degradation rate is 0.032% per cycle before 150th cycle, and after 150 cycles, it is 0.0188%, while after the battery has suffered 600 cycles, the capacity declines faster at 0.0271% per cycle. For 1.2C rate charging, the capacity degradation rate is 0.0472% per cycle before 100th cycle, and during 100th cycle to 400th cycle, it is 0.0226%, while the accelerated degradation rate is 0.0356% per cycle after 400 cycles. The case of 1.5C rate charging is distinct from other charging rates. The average capacity degradation rate of 1.5C is 0.078% per cycle, which is far greater than other rates.

As shown in Fig. 2 (a) and (b), the battery aging process under different charging rates is divided into three stages according to the capacity degradation rate. During the first stage, the new batteries start suffering cycle, the capacity degradation rate is relatively large and greatly affected by the charging current rates. This may be due to the activation of new battery and SEI film formation at the negative electrode surface during cycle, which will consume more active material inside the battery. During the second stage, the new batteries have suffered a certain number of cycles, the SEI film has been fully formed, and the internal states of the tested batteries become stable gradually. As a result, the capacity degradation rate decreases, and it is insensitive to the charging current rates in a certain range such as 1C. Therefore, this stage is suitable to increase charging current to reduce charging time. During the third stage, which is the later period of battery aging, due to the battery aging along with the loss of electrode active material, the charge acceptance capability of the graphite anode becomes weaker, and the capacity degradation rate is sensitive to charging current and the same charging current will accelerate battery degradation. It is better to reduce charging current in this stage to delay battery aging.

In Fig. 2 (c), the effects of different charging cut-off voltage limits on the maximum available capacity degradation are examined. The charging capacity will diminish as reducing the cut-off voltage, so in order to conduct reasonable comparison, the abscissa in Fig. 2 (c) is the through-put capacity during cycle. Capacity degradation will be exacerbated significantly when the cut-off voltage is set as 4.3 V.

Table 2

Cycle life tests methodology at different charging currents and charging cut-off voltages.

Cell number	Charging rate	Charging cut-off voltage/V
Cell 1, 2, 3	0.5C	4.2 V
Cell 4, 5, 6	1C	4.2 V
Cell 7, 8, 9	1C	4.1 V
Cell 10, 11, 12	0.8C	4.2 V
Cell 13, 14, 15	1C	4.3 V
Cell 16, 17, 18	1.2C	4.2 V
Cell 19, 20, 21	1.5C	4.2 V

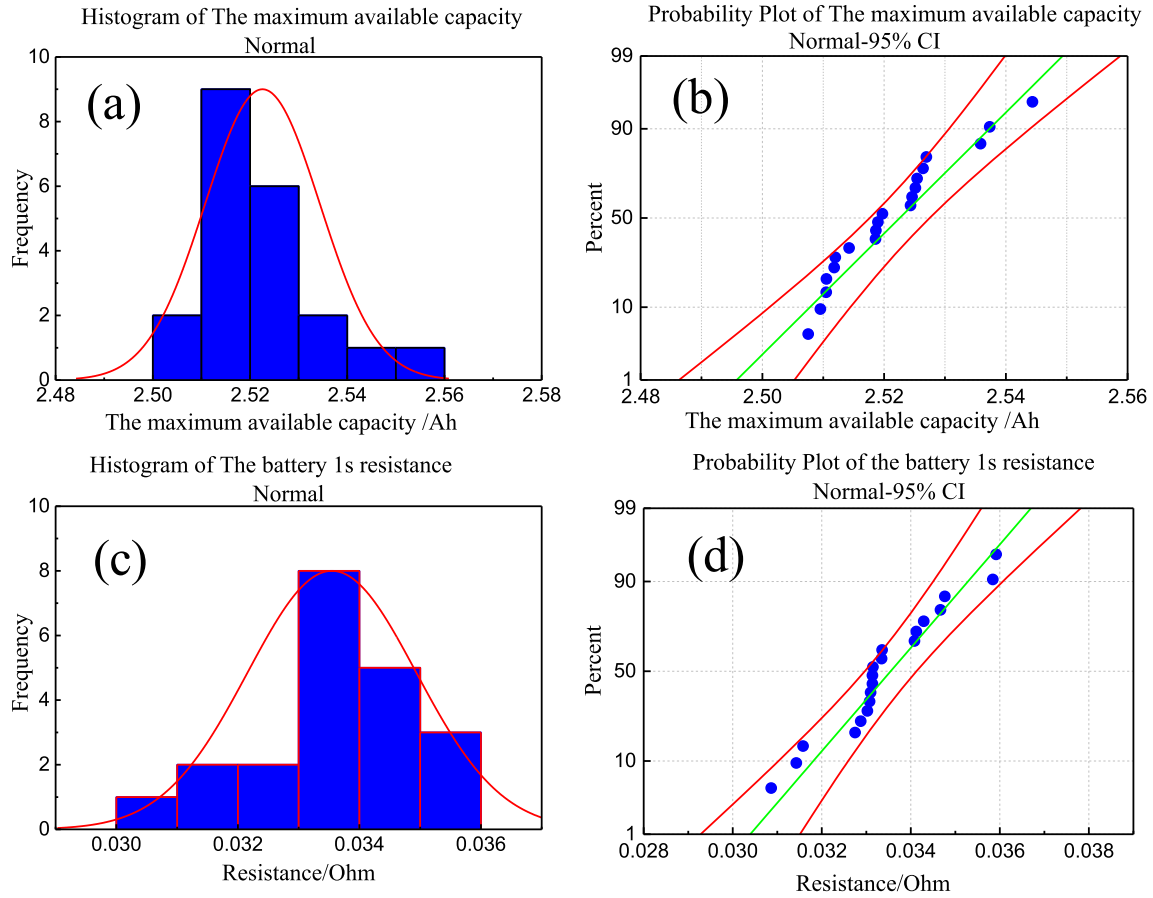


Fig. 1. (a) Initial capacity distribution. (b) Normal probability plot of the initial capacity distribution. (c) Initial resistance distribution. (d) Normal probability plot of the initial resistance distribution.

When the cut-off voltage decreases from 4.2 V–4.1 V, the capacity degradation rate will be reduced by 38% correspondingly from 0.0541% to 0.0338% per 10Ah through-put capacity. The results indicate that limiting the charging cut-off voltage within a secure range is significant for suppressing battery degradation.

3.3. Increase of battery internal resistance

The battery internal resistance comprising polarization resistance and ohmic resistance is the main parameter to characterize the kinetics performance of lithium-ion battery. The battery terminal voltage comes to an abrupt change while undergoing current mutation. The internal resistance is calculated by dividing the voltage change by the current change. The response time of ohmic resistance is very fast, almost instantaneous, followed by a polarization response after a few seconds. In this paper, we consider the 1s response resistance as ohmic resistance and 60s response resistance as the total internal resistance. The polarization resistance comes from the difference between the total resistance and ohmic resistance [18]. Fig. 3 (a) displays the increase of 1s resistance at 40% SOC along with battery aging under different charging current rates. To exclude the effect of random factors, the mean resistance of the three tested batteries is used for comparison. When the charging current rate is less than 1C, the increase rate of 1s resistance along with battery aging is almost the same. It is indicated that within a certain range, the charging current size has little effect on the increase of 1s resistance. However, when the charging current rate is greater than 1C, the increase rate of 1s

resistance will be accelerated rapidly as the increasing of charging current rate.

In order to rule out the effect of random factors on the results, 3 batteries have been put in each charging stress case. Table 3 shows the 1s resistance of 9 batteries at 0, 650, and 750 cycles under 0.5C, 0.8C and 1C charging cycle. The variance of 1s resistance among the three charging stress cases can be compared with the variance of 1s resistance in the same case to determine the effect of charging current rate on the increase of battery resistance. This is called the Analysis of Variance (ANOVA) in the statistical field [18].

In the following analysis, R is used to represent the battery 1s resistance. Formula (1) is used to calculate the variance of the battery 1s resistance among the three different cycle cases (S_I), n_i is the tested batteries number in cycle case i , $\bar{R}_i$ is the average resistance of the tested batteries in cycle case i , $\bar{R}$ is the average resistance of all tested batteries in the three cycle cases. The variance of battery resistance in the same cycle case (S_e) is calculated by Formula (2), in which R_{ij} is the 1s resistance of the tested battery j in cycle case i .

$$S_I = \sum_{i=1}^3 n_i (\bar{R}_i - \bar{R})^2 \quad (1)$$

$$S_e = \sum_{i=1}^3 \sum_{j=1}^3 (R_{ij} - \bar{R}_i)^2 \quad (2)$$

Mean squares (MS) is defined in Formula (3) to remove the effect

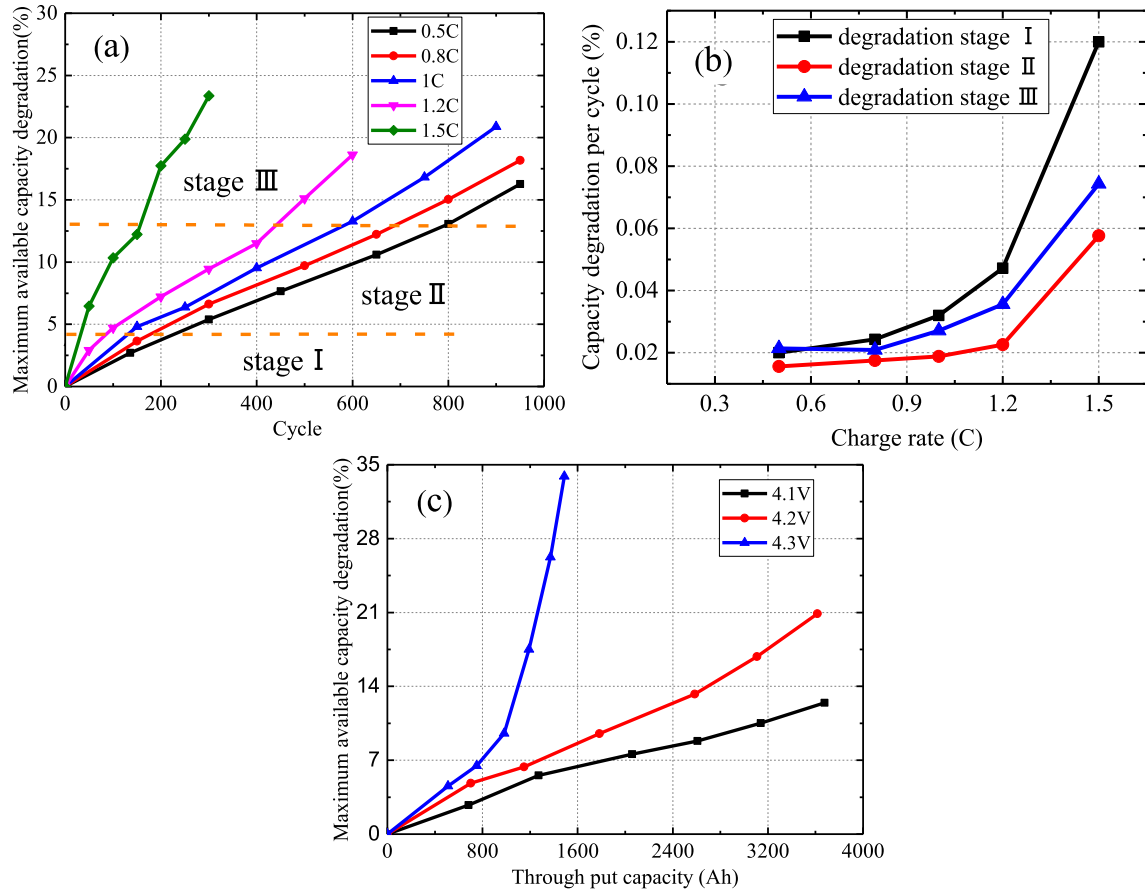


Fig. 2. (a) The maximum available capacity degradation under different charging rates. (b) The maximum available capacity degradation rate under different charging currents. (c) The maximum available capacity degradation under different charging cut-off voltages.

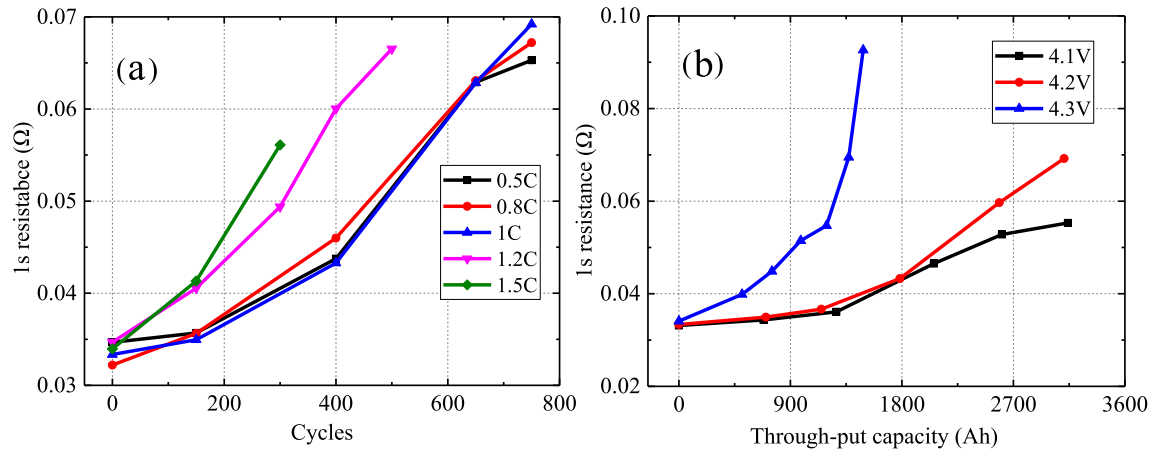


Fig. 3. 1s resistance increase along with the battery cycle aging at (a) different charging current rates (b) different charging cut-off voltages.

Table 3
1s resistance increase along with the battery aging under different charging current rates.

	cycles	0.5C			0.8C			1C			F
		Ce1	Ce2	Ce3	Ce10	Ce11	Ce12	Ce4	Ce5	Ce6	
R_{1s} (Ω)	0	0.0328	0.0347	0.0359	0.0309	0.0314	0.0343	0.0331	0.0330	0.0334	1.98
	650	0.0655	0.0629	0.0572	0.0631	0.0606	0.0650	0.0615	0.0593	0.0629	0.243
	750	0.0597	0.0653	0.0580	0.0672	0.0638	0.0687	0.0673	0.0688	0.0692	6.15

Table 4
1s resistance increase along with the battery aging under different cut-off voltages.

Through-put capacity		4.2 V			4.1 V			F
		Ce4	Ce5	Ce6	Ce7	Ce8	Ce9	
R_{1s} (Ω)	0 Ah	0.0331	0.0330	0.0334	0.0331	0.0331	0.0316	1.21
	1783Ah	0.0405	0.0418	0.0433	0.0431	0.0452	0.0421	1.72
	2607 Ah	0.0599	0.0552	0.0601	0.0495	0.0528	0.0540	9.08
	3110Ah	0.0653	0.0688	0.0692	0.0521	0.0553	0.0594	24.7

of sample number on the calculation, DOF is the degrees of freedom.

$$MS = S/DOF \quad (3)$$

Then, F value of Fisher test is calculated with Formula (4), in which DOF_I is equal to 2 and DOF_e is 6.

$$F = \frac{MS_I}{MS_e} = \frac{S_I/DOF_I}{S_e/DOF_e} \quad (4)$$

Finally, Formula (5) defines the F_I , which is the $1-\alpha$ quantile of F -distribution, and α is significance level and equal to 0.05. If F is smaller than F_I , the charging current rates (0.5C, 0.8C and 1C) is not statistically significant to the increase of battery 1s resistance along with battery cycle aging.

$$F_1 = F_{1-\alpha}(DOF_I, DOF_e) \quad (5)$$

The calculated F is shown in Table 3. At the 650th cycle, the F is smaller than F_I , which is equal to 5.14 in this situation, while the F at the 750th cycle is greater than F_I . It is indicated that before the 650th cycle, within a range of 1C, the charging current rate has no significant effect on 1s resistance increase, but after the 650th cycle, the batteries cycling at 1C rate charging suffer a much faster increase of 1s resistance. It may be caused by the weaker charge acceptance capability due to aging. Charging at a great current will accelerate the degradation of battery kinetics performance.

The increase of 1s resistance at 40% SOC along with battery aging under different charging cut-off voltages is illustrated at Fig. 3 (b). Table 4 shows the 1s resistance of 6 batteries and the corresponding F value at different cycle stages. When the cut-off voltage is 4.3 V, the 1s resistance increases rapidly, which indicates that increasing cut-off voltage will induce accelerated degradation of battery kinetics performance. When the cut-off voltage is 4.2 V or 4.1 V, the increase rate of 1s resistance is basically equal at the early stage of battery cycle. When the through-put capacity is 1783Ah, the F is smaller than F_I , which is equal to 7.71 in this situation, indicating that cycling at 4.1 V or 4.2 V has no significant effect on the increase of 1s resistance. However, when the battery degrades to a certain extent, the F is greater than F_I , and the resistance increase of the battery cycling at 4.2 V is faster than 4.1 V, which indicates that reducing the cut-off voltage at the later stage of battery cycle will delay the decline of battery kinetics performance.

Table 5 shows the polarization resistance of 9 batteries at

different aging stages under 0.5C, 0.8C and 1C charging cycle, and the corresponding F values are also listed in Table 5. The increase trend of polarization resistance along with battery aging is identical to 1s resistance at all the charging current rates. Fig. 4 compares the increase of polarization resistance along with battery aging under different charging cut-off voltages and each charging cut-off voltage condition has three tested batteries. When the cut-off voltage is 4.3 V, the polarization resistance increases rapidly. When the cut-off voltage is 4.2 V or 4.1 V, it has no significant effects on the increase of polarization resistance until the through-put capacity reaches to 3000Ah. After 3000Ah, cycling at 4.2 V will cause more polarization resistance increase.

4. Discussion

4.1. Incremental capacity curve peak area evolution analysis

In this section, incremental capacity analysis is used to deduce the battery aging mechanisms, which has been reported in several studies [28–31]. Incremental capacity (IC) curves (dQ/dV-V) are calculated by taking the derivative of voltage profile (Q-V curve), and it can reflect the evolution of electro-chemical procedure during the charging process. Peaks in the IC curve can reveal the phase transition phenomena in the battery, so the evolution of the peaks' area, position and shape can be used to identify battery aging mechanisms. Fig. 5 shows the evolution of IC curve along with battery aging, which is derivative from 0.05C charging process during RPT. There are three distinct peaks in the curve and the areas of the three peaks decrease gradually with the battery aging, which can reflect the loss of electrodes active materials and lithium inventory.

For LiCoO₂ cathode, it exhibits a pair of strong redox peak at 3.97/3.85 V and two very weak peaks at 4.06/4.03 V and 4.19/4.13 V, corresponding to the oxidation/reduction reactions of Co³⁺/Co⁴⁺, respectively [35]. For graphite anode, the phase transformation characteristics are shown in Ref. [36], the graphite undergoes five distinct phase transformations. The phase transformation of LiCoO₂ cathode shall be convoluted with those of the graphite anode, resulting in a series of distinct IC peaks. The dQ/dV curve in Fig. 5 primarily reflects the characteristics of the LiCoO₂ cathode and the graphite anode, which is similar to LiMn_{1/3}Ni_{1/3}Co_{1/3}O₂ cathode based battery, due to the approximate phase transformation characteristics of LiCoO₂ and NMC. Reference to Ref [15,36], the

Table 5
Polarization resistance increase with the battery aging under different charging current rates.

Cycles		0.5C			0.8C			1C			F
		Ce1	Ce2	Ce3	Ce10	Ce11	Ce12	Ce4	Ce5	Ce6	
R_p (Ω)	0	0.0243	0.0234	0.0226	0.0260	0.0238	0.0232	0.0252	0.0246	0.0212	0.26
	150	0.0273	0.0273	0.0257	0.0271	0.0277	0.0310	0.0284	0.0274	0.0287	1.44
	400	0.0239	0.0268	0.0292	0.0296	0.0297	0.0288	0.0282	0.0278	0.0280	2.29
	650	0.0342	0.0363	0.0299	0.0387	0.0389	0.0382	0.0369	0.0350	0.0398	3.82
	750	0.0339	0.0384	0.0327	0.0392	0.0397	0.0391	0.0422	0.0392	0.0438	7.19

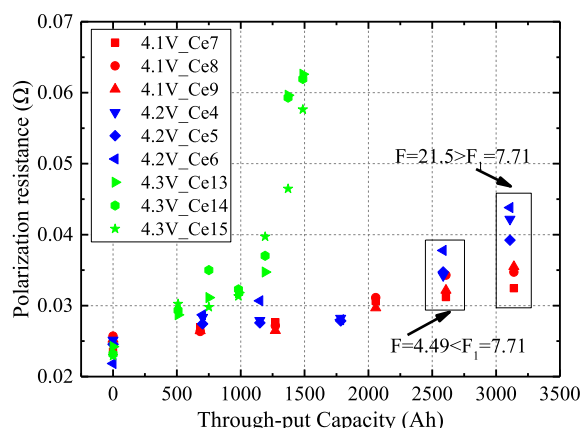


Fig. 4. Polarization resistance increase with battery aging under different charging cut-off voltages.

relationship between each peak and corresponding electro-chemical reaction mechanism is elaborated. Peak III reveals phase transformation of graphite anode and it can be used as status indicator for graphite anode. Peak II reveals the strong redox peak of LiCoO_2 materials and it is an indicator for LiCoO_2 cathode. Peak I corresponds to the electro-chemical procedure at the end of the charging process and it implies the extent of Li insertion in the graphite anode.

4.1.1. The evolution of IC curve peaks area along with cycle under different charging current rates

Fig. 6(a–d) illustrates the evolution of IC curve peaks area along with cycle under different charging current rates. Peak III is at the voltage platform of graphite anode, and its area evolution that is shown in Fig. 6 (a) can reflect the loss of negative electrode (NE) active materials (LAM_{NE}). The area of peak III decreases along with battery aging and then remains constant. It may be due to the battery activation and SEI film formation at the beginning of cycle, which will result in LAM_{NE} . After suffering a certain number of cycles, the internal states of the battery gradually become stable, and as a result, the LAM_{NE} will be slowed down. The area reduction of peak III under 0.5C rate charging cycle is significantly less than the other charging current rates. As for charging at 1.5C rate, the LAM_{NE} accelerates significantly, which may be caused by the shock

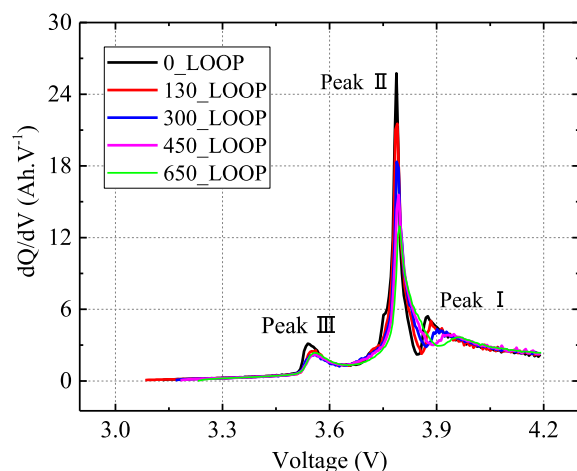


Fig. 5. The evolution of Increment capacity curve along with battery cycling at 0.5C charging current.

of high charging current to the NE. It is indicated that 1.5C rate exceeds the maximum charging current that the NE can withstand. When the charging current is 1C or 1.2C, the area of peak III starts to decrease again after 300 cycles, this is because that the charge acceptance capability gradually gets weak with battery aging, and 1.2C or 1C rate has exceeded battery's charge acceptance capability after 300 cycles. Thus, LAM_{NE} reappears. As for charging at 0.5C or 0.8C, the LAM_{NE} reappears when the battery undergoes 650 cycles.

Both peak III and peak II are at the voltage platform of positive electrode (PE) and they correspond to the phase transition of PE active materials. As shown in Fig. 6(b–c), the evolution of the two peaks' area can reflect the loss of positive electrode active materials (LAM_{PE}). For 0.5C rate charging, the area of peak II increases in the first 150 cycles, and then remains stable during the 150th cycle to 650th cycle. The total area of peak II and peak III almost does not reduce in the first 650 cycles, and it starts to reduce after 650 cycles, which indicates that there is almost no LAM_{PE} before 650 cycles and only when the battery degrades to a certain extent, the LAM_{PE} begins to appear. For 0.8C rate charging, the total area of peak II and peak III decreases at a steady rate along with battery aging, corresponding to the stable LAM_{PE} . For 1C rate charging, the total area of peak II and peak III decreases at almost the same rate as 0.8C before 250 cycles, but the reduction of peak area accelerates after 250 cycles, which indicates that the LAM_{PE} accelerates after 250 cycles. As for 1.2C and 1.5C charging, the total area of peak II and peak III decreases rapidly with the increase of the cycle number, especially for 1.5C, this is because that the shock of high charging current to PE results in rapid LAM_{PE} .

Peak I corresponds to the electro-chemical procedure at the end stage of the charging process, and its area suffers the effect of lithium inventory. As shown in Fig. 6 (d), for 0.5C rate charging, the total area of peak II and peak III almost does not reduce before 650 cycles, while the area of peak I decreases at a stable rate along with battery aging, which indicates that the capacity corresponding to peak I cannot be fully charged because of the loss of lithium inventory (LLI). Therefore, the LLI is the main cause to the battery capacity degradation. However, after 650 cycles, the reduction of peak I area slows down. The cause of this phenomenon may be that after 650 cycles, the electrodes active materials restart to reduce, and the electro-chemical procedures of peak II and peak III will consume less lithium inventory, thus more lithium inventory will be left to peak I. For 0.8C rate charging, the reduction rate of the peak I is stable but smaller than 0.5C. It is because that charging at 0.8C will cause more LAM than 0.5C. For 1C rate charging, the decrease of peak I is almost equal to 0.8C before 250 cycles. From previous analysis, we can know that for 1C charging, the total area of peak II and peak III also suffers the same loss as 0.8C. Therefore, there is no significant difference in the LLI between 0.8C and 1C. As for 1.2C and 1.5C rate charging, more electrodes active materials are lost, leaving more lithium inventory to peak I than 0.8C and 1C, but the reduction rate of the peak I is almost equal to 0.8C or 1C, which indicates that charging at 1.2C or 1.5C will cause more LLI than other rates. This phenomenon mainly stems from the fact that due to the limitation of the solid-phase diffusion rate of lithium inserted in graphite anode, a high charging current will cause the aggregation of lithium on the surface of graphite anode, thus side reactions such as lithium dendrites will be aggravated, and more lithium inventory will be consumed. Therefore, when the charging current rate is higher than 1C, the higher the charging current rate is, the more LLI is caused. However, when the charging current is lower than 1C, the difference in LLI is not significant.

4.1.2. The evolution of IC curve peaks area along with cycle under different cut-off voltages

Fig. 6(e–h) illustrates the evolution of IC curve peaks area along

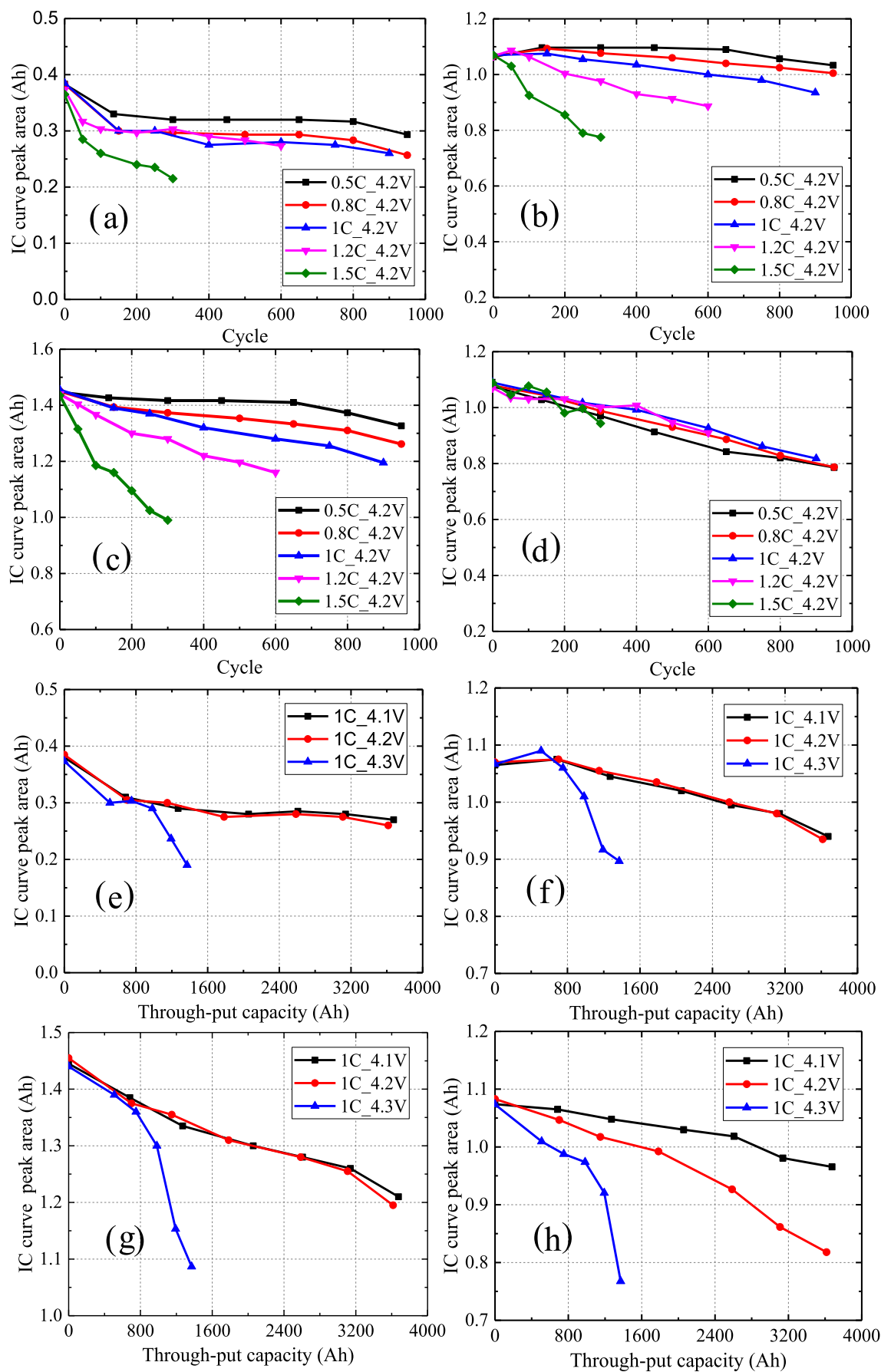


Fig. 6. The evolution of IC curve peaks area along with cycle under different charging current rates (a) peak III, (b) peak II, (c) peak III + II, (d) peak I. The evolution of IC curve peaks area along with cycle under different cut-off voltages (e) peak III, (f) peak II, (g) peak III + II, (h) peak I.

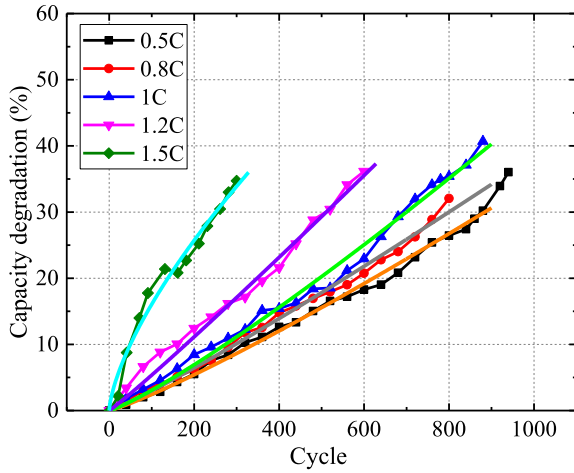


Fig. 7. The capacity degradation upon continuous cycle aging at different charging current rate.

with cycle under different cut-off voltages. When the cut-off voltage is 4.1 V, the area evolution of peak III and peak II is almost the same as 4.2 V, which demonstrates that reducing cut-off voltage cannot reduce the loss of electrodes active material. However, when increasing cut-off voltage to 4.3 V, the areas of peak III and peak II decrease quickly. In this situation, the PE will be over-charged and NE will be over discharged, which will cause great impact on the electrodes materials. Therefore, improving charging cut-off voltage will cause more LAM_{PE} and LAM_{NE} . When the cut-off voltage is 4.1 V, the area reduction of peak I is less than 4.2 V, which indicates that lowering cut-off voltage can reduce LLI. When the cut-off voltage is 4.2 V, the area reduction of peak I accelerates after the through-put capacity reaches to 1600Ah, demonstrating that the LLI accelerates. Therefore, when the battery degrades to a certain extent, it is better to lower the cut-off voltage appropriately to reduce LLI. When improving the cut-off voltage to 4.3 V, more LLI is observed than 4.2 V. In the last stage of charging process, the solid-phase diffusion capability of lithium inserted in graphite anode gets worse, resulting in the aggregation of lithium on the surface of graphite anode. So the higher the cut-off voltage is, the more LLI is caused.

4.2. Capacity degradation rate model

4.2.1. Capacity degradation rate model under different charging current rates

Fig. 7 displays the capacity degradation of the tested batteries upon continuous cycle aging at different charging current rates.

Formula (6) is the common used experimental formula that describes battery cycle life. In the Formula, K and α are the identification parameters that are related to current, cut-off voltage and temperature, and x is the battery cycle number or through-put capacity. To identify those parameters, the capacity degradation curves under different charging current rates are fitted by Formula (6). As illustrated in Fig. 7, the fitting results indicate that the formula can fit the decline trend of battery capacity accurately. Table 6 shows the estimated values of K and α under different charging currents. By deriving Formula (6), the capacity degradation rate (DS) is obtained in Formula (8), and then Formula (8) is transformed

Table 6

Estimated parameters K and α in Formula (6) at different charging current rates.

	0.5C	0.8C	1C	1.2C	1.5C
K	0.0121	0.0116	0.141	0.0414	0.7028
α	1.152	1.183	1.17	1.056	0.6795

to Formula (9) by replacing x with Formula (7) which is the inverse solution of Formula (6). By using Formula (9), the capacity degradation speed corresponding to aging states at different charging currents can be calculated and the results are shown in Fig. 8 (a). The degradation speed of 1.5C rate is obviously larger than other rates. Battery aging mechanisms at 1.5C rate charging are significantly different from other rates, which has been demonstrated section 4.1. Fig. 8 (b) displays the capacity degradation speed corresponding to charging currents at different aging states, which satisfies Formula (10). The curves in Fig. 8 (b) are fitted by Formula (10) to obtain the unknown parameters which are listed in Table 7. Through the above process, as shown in Formula (10), the capacity degradation speed model thinking of different charging current rates (I_c) and capacity degradation states ($Capacity_{loss}$) is established.

$$Capacity_{loss} = k \times x^\alpha \quad (6)$$

$$x = \left(\frac{Capacity_{loss}}{k} \right)^{1/\alpha} \quad (7)$$

$$DS = \frac{dCapacity_{loss}}{dx} = \alpha k \times x^{\alpha-1} \quad (8)$$

$$DS = \alpha k \times \left(\frac{Capacity_{loss}}{k} \right)^{\frac{\alpha-1}{\alpha}} = \alpha \times k^{\frac{1}{\alpha}} \times Capacity_{loss}^{\frac{\alpha-1}{\alpha}} \quad (9)$$

$$DS(Capacity_{loss}, I_c) = a(Capacity_{loss}) \times I_c^{b(Capacity_{loss})} + c(Capacity_{loss}) \quad (10)$$

4.2.2. Capacity degradation rate model under different cut-off voltages

Fig. 8 (c) shows the capacity degradation of the tested batteries upon continuous cycle aging at different cut-off voltages and the curves in Fig. 8 (c) are fitted by Formula (6), but in this situation, x is the through-put capacity. Table 8 shows the estimated parameters. When the cut-off voltage is 4.1 V or 4.2 V, the formula can fit the capacity degradation trend accurately, but as for 4.3 V, the fitting error is large in the early stage of degradation. By using Formula (6)–(9), the capacity degradation rate corresponding to aging states at different charging cut-off voltages can be calculated. As shown in Fig. 8 (d), the degradation rate at 4.3 V is far greater than 4.1 V and 4.2 V. Therefore, the charging cut-off voltage should not exceed 4.2 V. When the cut-off voltage is in the range of 4.1 V–4.2 V, the capacity degradation rate can be calculated by linear interpolation using Formula (11). Through the above process, as shown in Formula (11), the capacity degradation speed model thinking of different cut-off voltages (V_c) and capacity degradation states is established.

$$DS(Capacity_{loss}, V_c) = DS(Capacity_{loss}, 4.1) + \frac{DS(Capacity_{loss}, 4.2) - DS(Capacity_{loss}, 4.1)}{4.2 - 4.1} \times (V_c - 4.1) \quad (11)$$

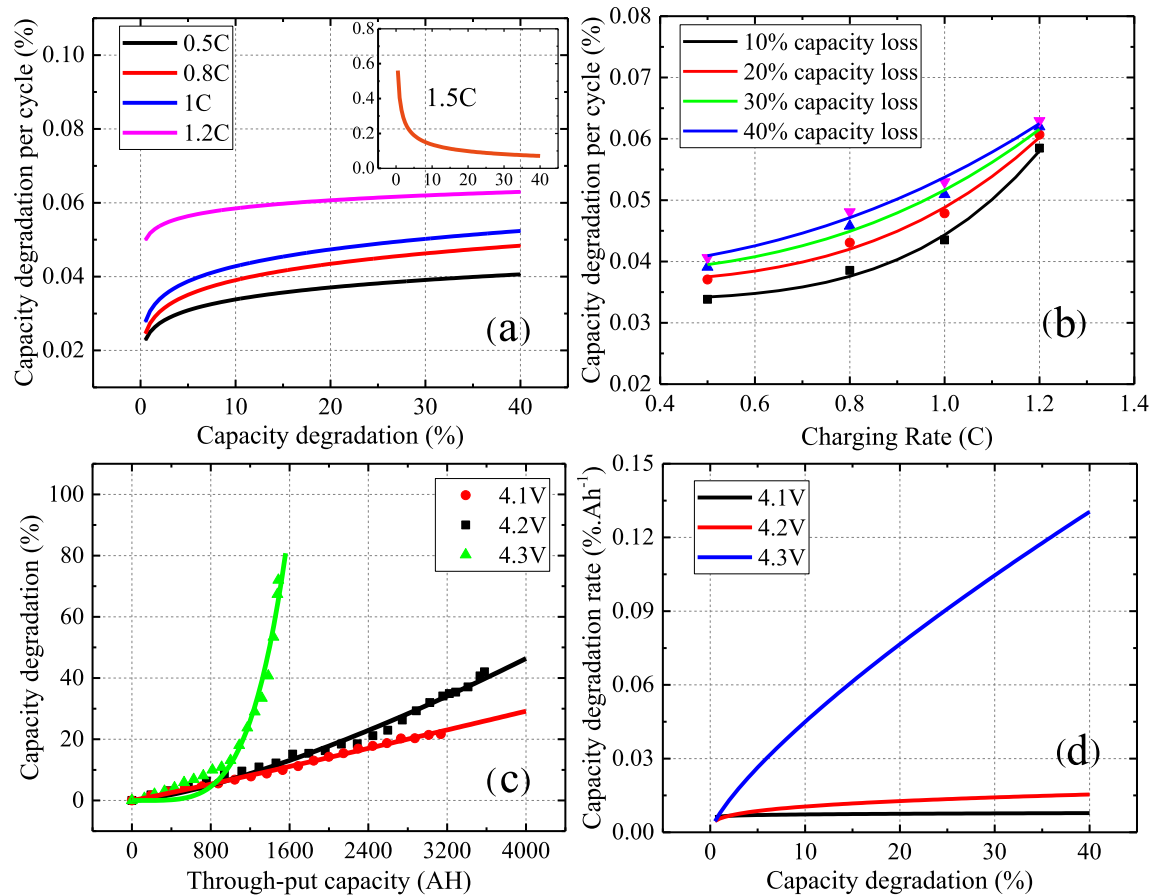


Fig. 8. (a) Capacity degradation speed at different battery aging states. (b) Capacity degradation speed at different charging current rates. (c) The capacity degradation of the tested batteries upon continuous cycle aging at different cut-off voltages. (d) Capacity degradation rate at different cut-off voltages.

4.3. The selection of charging current and cut-off voltage

From the previous analysis, we can know that there exist critical values for charging current rate and cut-off voltage. When the charging current or cut-off voltage exceed the critical values, the capacity degradation and resistance increase will be accelerated dramatically, and also more LAM and LLI will be observed. For the tested batteries, the critical charging current rate is 1C and critical charging cut-off voltage is 4.2 V. When the charging stresses are less than the critical values, with the increase of charging stresses, the acceleration of battery capacity degradation and resistance increase is not significant. Therefore, comprehensively considering charging time, charging capacity and battery lifespan, the reasonable selections of charging stresses are 1C and 4.2 V. However, after the battery suffers a certain number of cycles at 1C and 4.2 V, the charge acceptance capability of the battery gets weak, and as a result, the LAM and LLI will accelerate. So, reducing charging stresses in an appropriate aging stage is necessary to delay battery aging.

When the charging current is lower than the critical value, it is the deciding factor of LAM while its effect on the amount of LLI is

not significant, and reducing charging current can restrain LAM. On the contrary, when the charging cut-off voltage is lower than the critical value, it is the deciding factor of LLI while it has no effect on LAM, and reducing charging cut-off voltage can restrain LLI. The time when the LAM accelerates should be taken as the crisis to reduce charging current and the time when the LLI accelerates should be taken as the crisis to reduce charging cut-off voltage. As for the batteries in this paper, the acceleration phenomenon of LAM was observed when the batteries suffered 250 cycles at 1C charging current, and the acceleration phenomenon of LLI was observed when the through-put capacity of the batteries charged at 4.2 V reached to 1600Ah.

The charging current rate and cut-off voltage can be reduced according to the battery aging state to maintain constant cycle capacity degradation speed. By the quantitative relation between the capacity degradation speed and charging stresses, the appropriate charging current and cut-off voltage can be calculated to maintain constant cycle capacity degradation speed, which will be further studied in our future work. In the research of next stage, based on the battery aging mechanisms and capacity degradation rate model under different charging stresses, we will propose the

Table 7
Model parameters at different capacity degradation states.

	10%	20%	30%	40%
<i>a</i>	0.01058	0.01236	0.01398	0.01566
<i>b</i>	4.577	3.587	2.938	2.441
<i>c</i>	0.03375	0.0364	0.03766	0.03806

Table 8
Estimated parameters *K* and α in Formula (6) at different cut-off voltages.

	4.1 V	4.2 V	4.3 V
<i>K</i>	4.66×10^{-3}	4.92×10^{-4}	1.26×10^{-12}
α	1.054	1.381	4.324

calculation method of maximum charging current and cut-off voltage corresponding to battery degradation states in the whole service life to extend battery lifetime.

5. Conclusions

In this paper, the influence of charging stresses on battery life is systematically analyzed, and the capacity degradation rate model under different charge stresses is established. The main conclusions are summarized as follows:

- (1) There exists a critical charging current rate, which is 1C for the tested batteries. When the charging current exceeds the critical value, the capacity degradation, resistance increase, LAM_{NE} , LAM_{PE} and LLI will accelerate dramatically. When the charging current is less than the critical value, with the increase of charging current, the acceleration of capacity degradation and resistance increase is not significant, furthermore, the charging current is the deciding factor of LAM while its effect on LLI is not significant, reducing charging current can restrain LAM.
- (2) There exists a critical charging cut-off voltage, which is 4.2 V for the tested batteries. When the cut-off voltage exceeds 4.2 V, more capacity degradation, resistance increase, LAM_{NE} , LAM_{PE} and LLI will be observed along with battery cycle aging. While reducing the cut-off voltage from 4.2 V will retard capacity degradation and LLI, but it has no effects on LAM_{PE} and LAM_{NE} , and also has little effects on internal resistance increase in the early stage of battery life.
- (3) The LAM and LLI will accelerate along with battery aging, especially for the batteries charged at critical charging stresses. Reducing charging stresses in an appropriate stage is necessary to delay battery aging. The time when the LAM occurs acceleration is taken as the crisis to reduce charging current and the time when the LLI occurs acceleration is taken as the crisis to reduce charging cut-off voltage.
- (4) An experiential capacity degradation rate model is established. The model can accurately describe the relationship between capacity degradation rate and charging stresses at different aging states.

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